Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/26/2023
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:	Oct 31, 2021: All paid appointments on the grant ended. No new personnel added to the grant since then.		
COI Name (Institution):			

Grant/Contract No.:	NNX15AB42G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Parasitoid wasps are obligate parasites of their insect hosts. Most parasitoid wasps of Drosophila attack larval stages of fruit flies and consume the developing host, while growing inside its body. A normal immune response to wasp attack consists of the so-called encapsulation response in which dozens of differentiated blood cells or hemocytes migrate toward the egg to tightly encase, melanize and kill it. Parasitoid wasps produce factors in their venom to disrupt this immune reaction to ensure their success. In this study we examined the relative success of natural hosts and their parasites reared in space. We also assessed the effects of spaceflight on the immune physiology of Drosophila hosts and on parasite virulence. Two host strains, without and with inflammatory tumors, and two wasp species, differing in infection strategies were used in the experiment. Using physiological, cellular and molecular readouts, we found: (a) Flies without tumors and chronic inflammation were more sensitive to spaceflight conditions than tumor-bearing flies suffering from inflammation. Many essential genes in 'healthy' flies were downregulated. (b) Spaceflight spiked the expression of inflammation genes in fruit flies of both strains; space-travel enhanced tumor growth in tumor-bearing flies; larval progeny of these 'space' flies showed a higher tumor burden and were more strongly immune reactive than 'ground' controls. (c) Spaceflight did not significantly affect parasitic wasp development, emergence, or the expression of virulence genes; (d) Parasite mutants of one wasp species with defects in wing color, wing shape, and ovipositor morphology were identified and obtained. Pathogens and parasites pose a threat to human and animal health during spaceflight. Our studies reveal the unique and shared effects of spaceflight on the physiology and gene expression on distinct strains of Drosophila hosts. They show that metazoan parasites retain their infection strategies in space and highlight the resilience of parasite development in the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Parasitic wasps of Drosophila are a powerful and increasingly well-characterized model for understanding longstanding questions about natural host/parasite interactions, allowing molecular genetic examination of both innate immune mechanisms in fly hosts and virulence mechanisms of their wasp parasites. Distant relatives of these parasitic wasps are used for biocontrol of agricultural pests worldwide. The "Earth benefits" impact of our research are as follows: (a) RNA-Seq analysis of adult fruit flies (wild type and mutants developing inflammatory microtumors) will help in understanding the effects of spaceflight in humans and develop countermeasures for immune dysfunction. (b) RNA-Seq analysis of parasite transcriptomes in response to spaceflight with emphasis on virulence-related genes is providing novel insights into how endoparasites (those that develop within their hosts) react to the pressures as they develop in low Earth orbit. Whereas RNA-Seq-based gene expression analysis is now a standard experimental approach in biomedical research, the paucity of space-related research opportunities applied to model organisms have not allowed the potential of such 'omics' approaches to be realized toward spaceflight experiments. Our studies of this unique host-parasite system are among the first of their kind and will provide foundational data and resources for future studies. (c) In addition, the isolation of mutant wasp lines will advance our understating of wasp genomics, genetics, and host-parasite biology. Future model organism studies can help assess the hazardous effects of radiation on soma versus germline genomes and inspire ways to avert these risks with innovative shielding designs. The space environment is likely to have as yet unknown impacts on the animal immune function; refining our understanding of host-parasite systems with new genetic tools on Earth will broaden our understanding of immune function in space. (d) Four undergraduate students and one post-doctoral fellow were trained over last year. Two Federal Work Study students assisted and contributed to the project.
	Whereas previous spaceflight experiments in model organisms – such as the fruit fly, Drosophila melanogaster, and in humans – have shown that the immune system function in space is compromised, similar studies examining the immune system's interaction with natural pathogens and parasites in space have not been undertaken. With increased investments and opportunities in spaceflight, space travel by humans is becoming more frequent and will even be the norm in the years to come. In this context, it is important to ask how spaceflight affects natural pathogens and parasites in model organisms and in human hosts, especially if the hosts have dysfunctional immunity or are immune-compromised. To address this question, we examined the effects of spaceflight in Drosophila melanogaster infested with their natural parasitoid wasps that journeyed to-and-back from the International Space Station, aboard SpaceX-14, on a 34-day mission in the spring of 2018. Parasitoid wasps are obligate parasites of their insect hosts. Most parasitoid wasps of Drosophila attack larval stages of fruit flies and consume the developing host, while growing inside its body. A normal immune response to wasp attack consists of the so-called encapsulation response in which dozens of differentiated blood cells or hemocytes migrate toward the egg to tightly encase, melanize, and kill it. Parasitoid wasps produce factors in their venom to disrupt this immune reaction to ensure their success. The experimental design of our project consisted of two host strains, tumor-free flies with normal immunity (control flies), and mutant flies with chronic inflammatory tumors, due to hyperactive immunity (CI flies). Each of these hosts was infected by either of two wasp species (fly-host co-cultures), that differ in their infection strategies. One wasp species modifies the shapes of hemocytes, whereas the other kills hemocytes, to block encapsulation and gain control over the host. Hemocytes fight off wasp infections by making up the wasp capsules. Parasite-free flies (fly-only cultures) were also included in the experiments as controls. We examined (a) the relative success of hosts versus parasites, (b) the effects of spaceflight on the immune physiology of Drosophila hosts, and (c) the effects of spaceflight on parasite virulence itself. We also searched for the detrimental effects of space radiation on wasp morphology. Using physiological, cellular, and molecular readouts of the space-returned animals and their corresponding ground controls, we found:

Task Progress:	(a) Fewer than expected control flies returned to Earth, while the tumor-bearing CI flies succeeded in space just as well as ground controls. Thus, tumor-free control flies were more sensitive to spaceflight conditions than CI flies. This result is counter-intuitive and unexpected. Molecular analysis revealed that in the tumor-free control flies (but not in the mutant CI flies), many genes important for essential life functions were downregulated. Furthermore, the orthologs of many of these genes have disease relevance in humans. In both fly strains, genes encoding extracellular matrix (ECM) proteins or ECM-associated proteins (together, matrisome proteins), that contribute to cuticle and chorion structures, were strongly affected in space. (b) Spaceflight spiked the expression of inflammation genes in fruit flies of both control and CI strains; space travel enhanced tumor growth in tumor-bearing flies; larval progeny of these 'space' flies showed a higher tumor burden and were more strongly immune reactive than 'ground' controls. In addition, inflammation-related genes were activated in both strains and even though the control flies did not develop tumors, the tumor burden in the space-returned CI strain increased. (c) Surprisingly, spaceflight did not significantly affect parasitic wasp development, or their emergence. In addition, the effects on the expression of virulence genes was also not significant. In one of the two wasp species, virulence particles were seen normally distributed within the hosts. Together these results suggest that the effects of the space stressors on the parasite, that develops within the fly host, are quite modest. (d) Orbiting ~400 kilometer above Earth, life in the International Space Station is subjected to somewhat higher levels of space radiation than organisms on Earth are. In this research project, we identified two parasite mutants of one of the two wasp species. These mutants show defects in wing color, wing shape, and ovipositor morphology. To identify these mutants, we examined the grandsons of the male wasps born in space. A mutant that affects wing color (aurum, for golden wings, instead of the normal grey wings) breeds true, while the other mutant in which wing shape is affected (kona, angular wing shape relative to the normal round edge), cannot lay eggs in hosts due to structural defects in the ovipositor. These genetic variants are the only live mutant strains for this class of parasites and will be useful for research on host-parasite interaction studies on Earth. Pathogens and parasites pose a threat to human and animal health during spaceflight. Our studies reveal the unique and shared effects of spaceflight on the physiology and gene expression on distinct strains of Drosophila hosts. They show that metazoan parasites retain their infection strategies in space and highlight the resilience of parasite development in the space environment. Continued analysis of such systems, using model hosts and their various parasites, will help establish multigenerational studies and obtain a fine-grained understanding of the long-term effects of space on diverse organisms.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
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